A Low-Cost Vision-Based Autonomous Navigation System for Mobile Robots Using Deep Learning and Geometric-AI Perception

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Declaration

We do hereby declare that the research works presented in this thesis entitled, ”Thesis/Report Title” are the results of our work. We further declare that the thesis/report has been compiled and written by us, and no part of this thesis/report has been submitted elsewhere for the requirements of any degree, award, diploma, or any other purposes except for publications. The materials that are obtained from other sources are duly acknowledged in this thesis.

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Approval

I hereby declare that the research works presented in this thesis entitled, **"A Low-Cost Vision-Based Autonomous Navigation System for Mobile Robots Using Deep Learning and Geometric-AI Perception"**, are the outcome of original works carried out by Tarikul Islam (ID: 22235103242), Sultan Ahmmed (ID: 22235103227), Md Soykot Ahamed (ID: 22235103247), Ali Akbar Rakib (ID: 22235103237), and Suraia Islam (ID: 22235103235) under my supervision. No part of this thesis has been submitted elsewhere for any degree, award, or diploma except for publications. This dissertation meets the requirements and standards for the degree of Bachelor of Science in Computer Science and Engineering.

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Dedication

We would like to dedicate this research to our loving parents,

whose endless love, sacrifice, and unwavering support

have been our greatest strength throughout this journey.

To our respected teachers and mentors,

who guided us with patience and wisdom,

and inspired us to think beyond the ordinary.

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who stood by us through every challenge

and made this experience truly memorable.

And to every curious mind that dares to explore,

may this work serve as a small step forward

in the pursuit of knowledge and innovation.

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Abstract

Autonomous mobile robot navigation has traditionally relied on expensive sensor fusion techniques, including LiDAR and depth cameras, to achieve accurate obstacle detection and path planning. However, such approaches significantly increase system cost and limit accessibility in low-resource environments. This research proposes a cost-effective, vision-only autonomous navigation framework that integrates deep learning-based perception with geometric reasoning using the ground plane assumption. A pretrained deep learning model is utilized to detect obstacles from RGB camera input, while a novel ground-plane verification layer is introduced to estimate real-world distances by leveraging fixed camera height and tilt angle. This hybrid Geometric-AI approach addresses the scale ambiguity problem commonly found in standard monocular depth estimation models, enabling more reliable distance estimation without additional hardware. The system is implemented and evaluated within the ROS and Gazebo simulation environment under varying conditions, including low-light and dynamic scenarios. Performance is measured using key metrics such as collision rate, success rate, and time-to-goal. Experimental results demonstrate that the proposed method improves navigation accuracy and robustness while maintaining low computational cost and zero increase in hardware requirements. This research highlights the feasibility of deploying efficient and affordable vision-based navigation systems for real-world robotic applications.

Keywords: Autonomous Navigation, Mobile Robots, Deep Learning, Monocular Vision, Ground Plane Estimation, Obstacle Detection, ROS, Gazebo, Distance Estimation, Low-Cost Robotics.

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List of Abbreviations and Acronyms

AI	Artificial Intelligence
BUBT	Bangladesh University of Business and Technology
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CR	Collision Rate
CSE	Computer Science and Engineering
DRL	Deep Reinforcement Learning
GPU	Graphics Processing Unit
GPS	Global Positioning System
IBVS	Image-Based Visual Servoing
LiDAR	Light Detection and Ranging
LTS	Long-Term Support
MPC	Model Predictive Control
NSR	Navigation Success Rate
PPO	Proximal Policy Optimization
RGB	Red, Green, Blue
ROS	Robot Operating System
SLAM	Simultaneous Localization and Mapping
UAV	Unmanned Aerial Vehicle
VRAM	Video Random Access Memory

Contents

Chapter 1

1 Introduction

Autonomous mobile robot navigation is a key research area combining computer vision, deep learning, and robotics. As robots are widely used in industrial and service environments, the need for low-cost and reliable navigation systems has increased. Traditional methods rely on expensive sensors like LiDAR and depth cameras, which are not suitable for low-resource settings. This work proposes a lightweight, camera-only navigation system using deep learning. The system is developed and tested in ROS and Gazebo simulation for reliable and repeatable evaluation.

1.1 Problem Statement

A comprehensive review of the existing literature reveals several major research gaps in camera-only autonomous navigation. Most existing systems do not evaluate performance under varying lighting conditions [1, 6, 7], limiting real-world deployment. Many approaches depend on multiple cameras or additional sensors such as RGB-D or LiDAR [2, 5], increasing cost and complexity. Furthermore, the majority of systems are tested only in static environments, with limited evaluation of dynamic obstacles [4, 3, 13]. Safety and computational efficiency are rarely balanced within a single framework [1, 12], and most methods rely on structured or predefined settings such as pre-built maps or fiducial markers [6, 8, 13]. Finally, no consistent benchmarking framework exists across varying lighting and obstacle conditions [14, 6].

To address these gaps, this project proposes a lightweight, RGB-only deep learning navigation system implemented within a ROS–Gazebo simulation environment and evaluated using collision rate and navigation success rate across multiple environmental conditions.

1.2 Objectives

The primary aim of this research is to design, implement, and evaluate a lightweight deep learning-based vision-only autonomous navigation framework for mobile robots. The specific objectives are:

- To review and critically analyze existing deep learning-based visual navigation and obstacle avoidance approaches, identifying their methodological strengths and limitations with respect to sensor requirements, computational cost, and robustness.
- To design a lightweight CNN or DRL-based navigation architecture that processes only monocular RGB camera input, without reliance on LiDAR, depth cameras, or GPS.
- To implement the proposed framework within the ROS middleware and deploy it in a Gazebo simulation environment configured with varied static and dynamic obstacle layouts.
- To evaluate system performance under three distinct environmental conditions, standard lighting, reduced lighting, and dynamic obstacle scenarios, using collision rate and navigation success rate as primary metrics.
- To analyze experimental results, identify residual limitations, and propose concrete directions for future improvement and potential sim-to-real transfer.

1.3 Motivations

This research is motivated by the convergence of three trends in modern robotics and artificial intelligence. First, the proliferation of low-cost embedded computing platforms (e.g., NVIDIA Jetson Nano, Raspberry Pi 5) has made it technically feasible to deploy neural network inference on small, mobile robot platforms, but only if the navigation model is sufficiently lightweight to run in real time on constrained hardware.

Second, the economic imperative for affordable autonomous systems is acute in developing regions, where labour automation is increasingly necessary but LiDAR-equipped robots remain prohibitively expensive. A camera-only navigation solution could democratize autonomous robotics by reducing per-robot sensor costs from thousands of dollars to tens of dollars.

Third, the maturation of physics-based robot simulators, particularly Gazebo integrated with ROS, has reduced the barrier to rigorous navigation research. Simulation-trained policies have recently been shown to surpass real-data-trained policies in success rate, validating the simulation-first research paradigm adopted in this work. These converging factors create a compelling opportunity and practical imperative to develop and validate a robust, low-cost, vision-only navigation solution.

1.4 Significance of the Research

This research contributes to the growing body of knowledge at the intersection of computer vision, deep learning, and mobile robotics. Its significance can be understood at three levels.

At the **scientific level**, this work provides a systematic, metric-driven comparison of vision-only navigation performance across environmental conditions within a reproducible ROS-Gazebo simulation framework. The findings will advance understanding of the conditions under which RGB-only deep learning navigation is viable and where it breaks down.

At the **engineering level**, the proposed lightweight architecture demonstrates that acceptable navigation performance, competitive with sensor-fused baselines, is achievable on cost-constrained hardware. This provides a blueprint for low-cost robot developers and startups in resource-limited economies.

At the **societal level**, affordable autonomous navigation enables applications that can improve quality of life: last-mile delivery in dense urban settings, assistive robots for elderly care, and autonomous warehouse logistics in small-to-medium enterprises. By reducing the sensor cost barrier, this research helps extend the benefits of autonomous robotics to a broader segment of society.

1.5 Research Contribution

The specific contributions of this project are as follows:

- A comprehensive critical review of recent vision-based autonomous navigation research, analysing existing methods in terms of sensor dependency, computational complexity, robustness, and evaluation limitations.
- A lightweight deep learning-based navigation framework (CNN/DRL) that operates exclusively on monocular RGB input, eliminating the need for LiDAR, depth cameras, or GPS, and is designed for cost-effective deployment.
- A complete simulation-based implementation using ROS and Gazebo, enabling a reproducible and controlled environment for testing autonomous navigation in diverse scenarios.
- A structured evaluation framework that defines three key test conditions, normal lighting, degraded lighting, and dynamic obstacle environments, using standardised metrics such as collision rate and navigation success rate.
- A focused analysis of lighting robustness and environmental variability, addressing a critical limitation in existing vision-based navigation systems.
- An investigation of sim-to-real transfer challenges, including identification of performance gaps and potential strategies for deploying the proposed system on real-world robotic platforms.

1.6 Complex Engineering Analysis

1.6.1 Mapping with Complex Engineering Problems (WP)

1. **WP1 (Depth of Knowledge Required):** The project requires integrated knowledge from multiple domains, including computer vision, deep learning, reinforcement learning, and robotics systems. Understanding RGB-based perception, navigation policies, and real-time system constraints makes it a multidisciplinary and knowledge-intensive problem.

2. **WP2 (Conflicting Requirements):** The system must balance multiple trade-offs such as navigation accuracy vs. computational efficiency, safety vs. speed, and simulation performance vs. real-world generalization. Managing these competing constraints makes the design complex.

3. **WP3 (Depth of Analysis Required):** The navigation system must be evaluated under multiple environmental conditions, including different lighting levels and dynamic obstacles. This requires systematic experimentation and statistical analysis using metrics like collision rate and navigation success rate.

4. **WP4 (Familiarity of Issues):** While vision-based navigation and DRL are established research areas, combining camera-only navigation with lighting robustness and simulation-based evaluation is still under-explored, making this a partially unfamiliar problem.

5. **WP6 (Stakeholder Involvement):** The system must meet the needs of robotics developers, hardware manufacturers, and end-users, particularly in low-cost deployment scenarios. It must ensure compatibility, reliability, and usability across different applications.

6. **WP7 (Interdependence):** The system components perception, depth estimation (if used), policy learning, and control are highly interdependent. Errors in perception directly affect navigation decisions, requiring the entire pipeline to be designed as a tightly integrated system.

1.6.2 Mapping with Complex Engineering Activities (EA)

1. **EA1 (Range of Resources):** The project utilizes multiple resources, including deep learning frameworks (PyTorch), DRL libraries (Stable-Baselines3), ROS middleware, Gazebo simulation, and GPU-based training environments.

2. **EA2 (Level of Interactions):** It involves complex interaction between perception models, DRL policy networks, ROS communication nodes, and the Gazebo simulation engine, requiring efficient real-time data flow and system coordination.

3. **EA3 (Innovation):** The project introduces a lightweight camera-only navigation framework evaluated across multiple environmental conditions, with a focus on lighting robustness and low-cost deployment, which is not widely explored in existing literature.

4. **EA4 (Consequences to Society and Environment):** The system can reduce the cost of autonomous navigation, enabling wider adoption in areas such as delivery systems, service robots, and assistive technologies, especially in developing regions.

5. **EA5 (Familiarity of Issues):** The project extends existing research in vision-based navigation by addressing real-world challenges such as lighting variation and dynamic environments within a simulation-first framework.

Chapter 2

2 Background Study

This chapter reviews research related to camera-only autonomous obstacle detection and navigation systems operating under varying lighting conditions in both indoor and outdoor environments. It summarizes key methods, contributions, and limitations in vision-based navigation, depth estimation, and simulation approaches. The review highlights research gaps in handling illumination changes and depth uncertainty, and outlines the problem statement and improvements targeted by the proposed system.

2.1 Literature Review

Sharma et al. (2025) [1] proposed MonoMPC, a monocular navigation system using a learned probabilistic collision model with risk-aware MPC planning. It reduces collision rates and improves navigation performance compared to ROS-based methods. However, the MPC planner is computationally expensive, and performance under different lighting conditions or outdoor environments is not evaluated.

Finke et al. (2026) [2] proposed a teacher–student framework where a LiDAR-based policy is distilled into a monocular depth-based student using PPO. The student outperforms the LiDAR teacher, especially for 3D obstacles. However, it requires multiple cameras, introduces latency due to depth estimation, and is only tested in structured environments.

Kim et al. (2025) [3] proposed CARE, a modular collision avoidance system that adds a repulsive force mechanism to visual navigation models. It improves safety and reduces collisions across multiple robots. However, it depends on the base policy and may struggle in narrow spaces, with evaluation limited to indoor environments.

Ding et al. (2023) [4] proposed a monocular camera-based DRL navigation system using a CNN to directly predict actions from RGB images. It performs well in complex obstacle scenarios and shows good sim-to-real transfer. However, it requires extensive training, struggles with reflective surfaces, and lacks evaluation in dynamic environments.

Lin et al. (2024) [5] proposed a vision-based SLAM navigation system combined with A* path planning for lightweight autonomous cars. It works efficiently on CPU-only systems and provides reliable obstacle avoidance. However, it depends on RGB-D sensors, and SLAM performance drops in dynamic or low-texture environments.

Suomela et al. (2025) [6] proposed FAINT, a simulation-trained Transformer-based navigation model that outperforms real-data-trained methods in real-world tests. It validates the effectiveness of simulation-based training. However, it requires predefined maps and does not address lighting variation or dynamic obstacles.

Dang et al. (2023) [7] proposed IRDC-Net, a lightweight semantic segmentation model for monocular navigation in indoor corridors. It achieves real-time performance with low computation. However, it is limited to structured indoor environments and does not handle dynamic obstacles or varying lighting conditions.

Wang et al. (2025) [8] proposed an RGB-only UAV navigation system using IBVS and monocular depth estimation for obstacle avoidance. It works without GPS or external sensors. However, it is limited to UAVs, depends on visible targets, and suffers from latency and poor performance in textureless environments.

Xie et al. (2017) [12] proposed a dueling deep Q-network (D3QN) approach for monocular vision-based obstacle avoidance without any depth sensor. The method demonstrates that RGB-only input is sufficient for basic obstacle avoidance using DRL. However, it lacks integration with modern depth estimation pipelines and is not evaluated under varying lighting conditions.

Eitel et al. (2021) [14] proposed an end-to-end DRL navigation system for real-world indoor environments using domain randomisation and fine-tuning on real images. It achieves high goal-reaching success rates on a physical robot. However, illumination robustness and dynamic obstacle handling are not systematically evaluated.

Pire et al. (2025) [13] proposed a ROS–Gazebo simulation framework for autonomous navigation of omnidirectional robots using AprilTag-based visual mapping and A* global planning. It provides a realistic and modular simulation testbed. However, it relies on fiducial markers and evaluates only static obstacle layouts without considering lighting variation.

2.2 Summary of Reviewed Literature

Table 1 summarizes the reviewed studies, including their primary research focus, methodology, key results, and identified limitations.

Table 1: Related Works in Vision-Based Navigation

Ref.	**Author(s)**	**Focus Area**	**Methodology**	**Key Result**	**Key Limitation**
[1]	Sharma et al. (2025)	Camera-only safe navigation	RGB $\rightarrow$ Depth $\rightarrow$ Learned collision model + MPC	Fewer collisions, better navigation	MPC is slow; lighting not tested

Ref.	Author(s)	Focus Area	Methodology	Key Result	Key Limitation
[2]	Finke et al. (2026)	Camera vs LiDAR navigation	Teacher (LiDAR) → Student (Depth + PPO)	Camera model beats LiDAR in 3D obstacles	Needs 4 cameras; latency
[3]	Kim et al. (2025)	Safe visual navigation	RGB → Repulsive force module (CARE)	Reduces collisions in new environments	Weak in narrow spaces; indoor only
[4]	Ding et al. (2023)	End-to-end camera navigation	CNN + DRL (RGB → action)	Works well for complex obstacles	Needs lots of training; no dynamic test
[5]	Lin et al. (2024)	SLAM-based navigation	Visual SLAM + A* planning	Works on CPU; reliable paths	Needs RGB-D; weak in dynamic scenes
[6]	Suomela et al. (2025)	Simulation-based navigation	Transformer trained in simulation (FAINT)	Beats real-data models	Needs map; lighting not tested
[7]	Dang et al. (2023)	Lightweight perception	MobileNetV2 + FCN segmentation	Fast and efficient	Only indoor; no lighting handling
[8]	Wang et al. (2025)	Camera-only UAV navigation	IBVS + Depth (MiDaS)	Works without GPS/LiDAR	Slow; needs visible targets
[12]	Xie et al. (2017)	Monocular DRL obstacle avoidance	D3QN (RGB → action)	RGB-only avoidance feasible	No depth pipeline; lighting not tested
[14]	Eitel et al. (2021)	Real-world indoor DRL navigation	End-to-end DRL + domain randomisation	86.7% goal-reaching on real robot	No lighting or dynamic evaluation

Ref.	Author(s)	Focus Area	Methodology	Key Result	Key Limitation
[13]	Pire et al. (2025)	ROS–Gazebo simulation framework	AprilTag mapping + A* planning	Modular, realistic simulation	Needs markers; static obstacles only

2.3 Conclusion

In summary, the research gap addressed by this project is the absence of a lightweight, RGB-only deep learning navigation framework that: (i) is evaluated under varied lighting and dynamic obstacle conditions; (ii) is benchmarked using both collision rate and success rate; and (iii) is implemented within a reproducible ROS–Gazebo simulation pipeline. The proposed framework described in Chapter 3 directly addresses each of these gaps.

Chapter 3

3 Methodology

This chapter presents the proposed methodology for a monocular vision-based autonomous navigation system capable of operating under varied lighting conditions and dynamic obstacle scenarios. The system relies solely on a single RGB camera and processes visual input through a multi-stage pipeline encompassing image preprocessing, monocular depth estimation, obstacle analysis, environment representation, path planning, navigation decision-making, and closed-loop feedback control. Figure 3.1 illustrates the complete proposed pipeline.

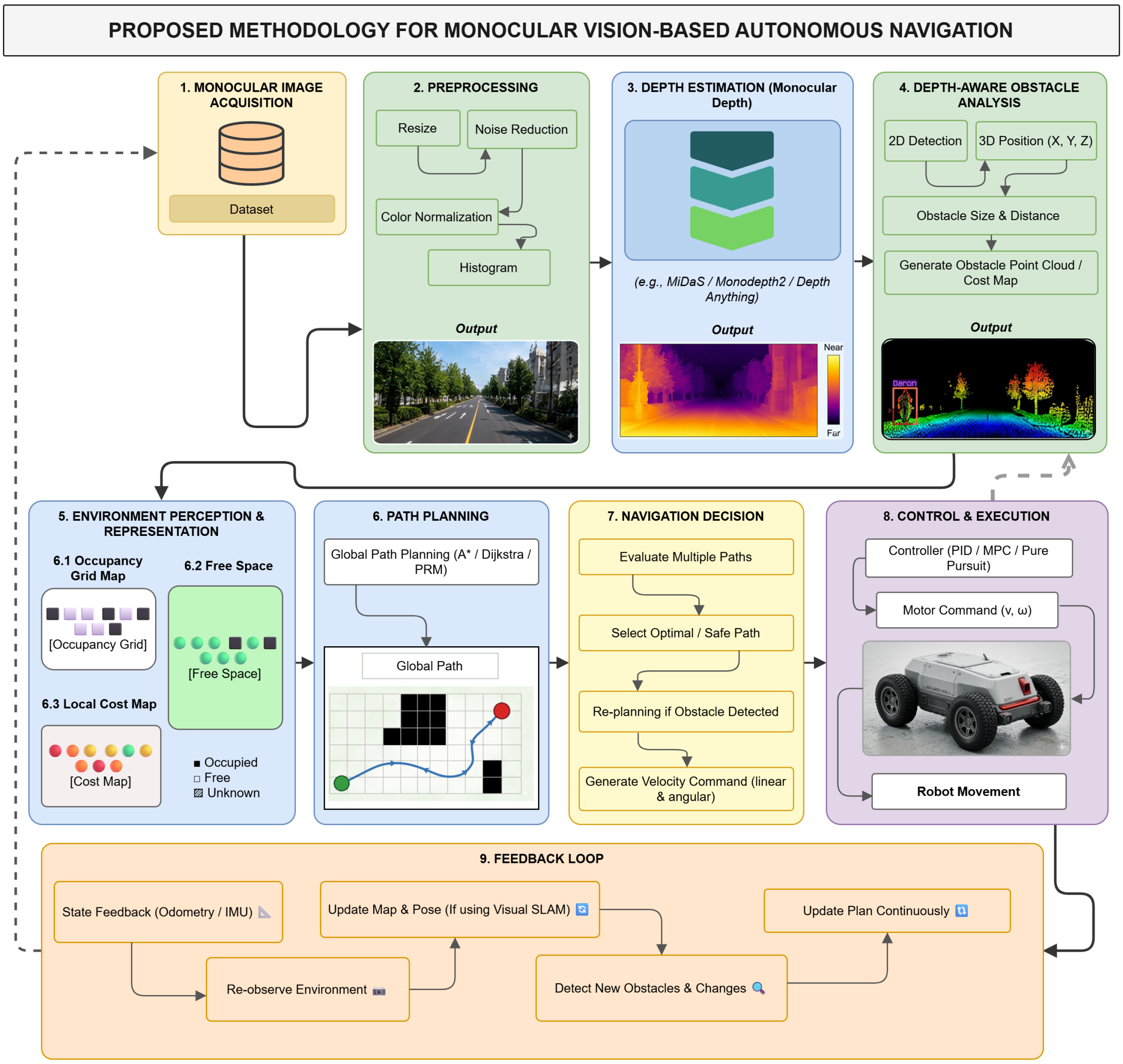

Figure 3.1: Proposed Methodology for Monocular Vision-Based Autonomous Navigation

3.1 System Overview

The proposed framework consists of ten interconnected stages, as illustrated in Figure 3.1. A monocular camera continuously captures RGB frames from the environment. These frames are preprocessed to normalise illumination and reduce noise before being passed to a pre-trained monocular depth estimation model. The resulting depth map, combined with a 2D object detection module, enables depth-aware obstacle analysis that generates a 3D cost map of the surroundings. The cost map feeds into an environment representation layer that constructs an occupancy grid and free-space map. A global path planner then computes an optimal trajectory, which is subsequently evaluated and refined by a navigation decision module. Finally, a low-level controller translates velocity commands into motor actions, and a feedback loop continuously updates the system state using odometry and sensor data.

3.2 Stage 1: Monocular Image Acquisition

The primary input to the system is a stream of monocular RGB images captured by a single forward-facing camera. In the simulation environment, image frames are collected from a Gazebo-based world using a camera plugin attached to the robot model. For training and evaluation purposes, publicly available autonomous driving and indoor navigation datasets are also employed to augment system robustness. The image stream serves as the sole perceptual input, making the system fully independent of LiDAR, stereo cameras, or GPS.

3.3 Stage 2: Image Preprocessing

Raw RGB frames are preprocessed before being passed to downstream modules to improve consistency and reduce the influence of environmental noise. The preprocessing pipeline consists of the following operations:

- **Resizing:** Input frames are resized to a fixed resolution compatible with the depth estimation model's expected input dimensions.
- **Noise Reduction:** Gaussian blurring and median filtering are applied to suppress image noise caused by camera sensor artefacts and low-light conditions.
- **Colour Normalisation:** Per-channel mean subtraction and variance scaling are applied to standardise pixel intensity distributions across different lighting environments.
- **Histogram Equalisation:** Adaptive histogram equalisation (CLAHE) is applied to enhance local contrast, particularly in underexposed or overexposed regions, improving performance under varying illumination.

The output of this stage is a normalised, denoised RGB image ready for depth inference and object detection.

3.4 Monocular Depth Estimation

A pre-trained monocular depth estimation model is used to infer a per-pixel depth map from each pre-processed RGB frame. The system evaluates and supports multiple state-of-the-art models, including MiDaS [9], Monodepth2 [10], and Depth Anything [11]. These models produce a relative or metric depth map encoding the estimated distance of each pixel from the camera plane.

The depth estimation stage is the core perceptual component of the proposed system. Since no stereo baseline or active depth sensor is available, the quality and generalisation ability of the monocular depth model directly determines the accuracy of downstream obstacle detection and navigation decisions. Models are selected and evaluated based on their inference latency, accuracy on unseen environments, and robustness under illumination changes.

3.5 Stage 4: (Implicit) Object Detection

In parallel with depth estimation, a lightweight 2D object detection module processes the RGB frame to identify the bounding boxes and class labels of obstacles in the scene. This detection output is fused with the depth map in the subsequent stage to associate each detected object with its estimated 3D position. The detection module is designed to operate at real-time frame rates, prioritising low latency over detection completeness to maintain system responsiveness.

3.6 Stage 5: Depth-Aware Obstacle Analysis

The outputs of the depth estimation and object detection stages are fused in this module to perform a comprehensive 3D analysis of detected obstacles. The following sub-steps are executed:

1. **3D Position Estimation:** Depth values within each bounding box are sampled and projected into 3D space using the camera intrinsic parameters, yielding (X,Y,Z) coordinates for each obstacle.

2. **Obstacle Size and Distance Computation:** The physical size and distance-to-contact of each obstacle are estimated from the projected 3D bounding volume.

3. **Obstacle Point Cloud Generation:** Detected obstacle regions are converted into a sparse 3D point cloud representation, encoding spatial extent and proximity.

4. **Cost Map Construction:** The point cloud is projected onto a 2D grid to generate a local cost map, where higher values indicate proximity to obstacles and represent traversal risk.

The output of this stage is a metric-scale cost map that encodes obstacle density and distance in the robot's navigable plane.

3.7 Stage 6: Environment Perception and Representation

The cost map generated in Stage 5 is used to construct a structured representation of the robot's immediate environment. Three complementary representations are maintained:

- **6.1 Occupancy Grid Map:** A discrete binary grid in which each cell is labelled as *Occupied*, *Free*, or *Unknown* based on the projected cost map values and accumulated sensor history.
- **6.2 Free Space Map:** A continuous representation of navigable areas derived from the occupancy grid, used to identify traversable corridors for path planning.
- **6.3 Local Cost Map:** A weighted spatial cost map that penalises cells near detected obstacles, enabling gradient-based path evaluation and safe trajectory selection.

These representations are maintained in a rolling window around the robot's current pose and updated at each perception cycle.

3.8 Stage 7: Path Planning

Given the occupancy and cost map representations, a global path planner computes a collision-free trajectory from the robot's current pose to the designated goal position. The system supports multiple classical planning algorithms, including:

- **A* Search:** Heuristic-guided graph search providing optimal paths with respect to a cost function.
- **Dijkstra's Algorithm:** Uniform-cost search ensuring shortest-path optimality without a heuristic.
- **Probabilistic Roadmap (PRM):** Sampling-based planning suited for high-dimensional or complex environments.

The planner outputs a sequence of waypoints forming a global path from start to goal. This path is passed to the navigation decision module for further refinement and local adaptation.

3.9 Stage 8: Navigation Decision

The navigation decision module evaluates the global path in the context of real-time obstacle updates and selects the safest and most efficient trajectory for immediate execution. The module operates through the following decision logic:

1. **Path Evaluation:** Multiple candidate paths derived from the global plan are scored using the local cost map and proximity-to-obstacle penalties.

2. **Optimal Path Selection:** The path with the lowest cumulative cost is selected for execution.

3. **Dynamic Re-planning:** If a newly detected obstacle intersects the current path, re-planning is triggered using the updated occupancy grid.

4. **Velocity Command Generation:** Linear (v) and angular (ω) velocity commands are computed from the selected path segment and passed to the control module.

This stage enables the robot to respond adaptively to dynamic environmental changes without halting navigation.

3.10 Stage 9: Control and Execution

The velocity commands produced by the navigation decision module are executed by a low-level robot controller. The system supports the following control strategies:

- **PID Controller:** A proportional–integral–derivative controller for smooth velocity tracking.

- **Model Predictive Control (MPC):** A receding-horizon controller that optimises control inputs over a short future window, offering improved smoothness and constraint satisfaction.

- **Pure Pursuit:** A geometric tracking algorithm that computes steering commands to follow a path through a lookahead point.

The selected controller translates the velocity command (v, ω) into differential drive motor signals, which are published to the robot's actuators via the ROS control interface. The resulting robot movement alters the environmental observations and feeds directly back into the perception pipeline.

3.11 Stage 10: Feedback Loop

A closed-loop feedback mechanism ensures continuous system adaptation throughout operation. The feedback loop performs the following functions:

- **State Feedback:** Odometry and IMU data are used to estimate the robot's current pose, velocity, and orientation at each time step.

- **Environment Re-observation:** The updated robot pose triggers a new monocular image capture and re-executes the perception pipeline, enabling continuous environment monitoring.

- **Map and Pose Update:** If a Visual SLAM module is active, the occupancy grid and robot pose estimate are updated using visual loop closure and keyframe data.
- **Obstacle Change Detection:** Newly appeared or relocated obstacles detected in the latest cost map are flagged and propagated to the navigation decision module.
- **Continuous Plan Update:** The global path and local trajectory are re-evaluated at each feedback cycle to maintain temporal consistency with the evolving environment.

This feedback architecture ensures that the robot responds in real time to dynamic obstacles, drift in pose estimation, and changing environmental conditions.

3.12 Summary of Proposed Pipeline

Table 2 summarises the ten stages of the proposed methodology, including the primary input, processing component, and output at each stage.

Table 2: Proposed Methodology Pipeline Stages

Stage	**Component**	**Input**	**Processing**	**Output**
1	Image Acquisition	Environment / Dataset	Single RGB camera capture	Raw RGB frame
2	Preprocessing	Raw RGB frame	Resize, denoise, normalise, CLAHE	Normalised RGB frame
3	Depth Estimation	Normalised frame	MiDaS / Monodepth2 / Depth Anything	Per-pixel depth map
4	Object Detection	Normalised frame	Lightweight 2D detector	Bounding boxes + labels
5	Obstacle Analysis	Depth map + detections	3D projection, point cloud, cost map	2D local cost map
6	Env. Representation	Cost map	Occupancy grid, free space, cost map	Structured env. map

Stage	Component	Input	Processing	Output
7	Path Planning	Env. map + goal pose	A* / Dijkstra / PRM	Global waypoint path
8	Navigation Decision	Global path + cost map	Path evaluation + re-planning	Velocity command (v, ω)
9	Control & Execution	(v, ω) command	PID / MPC / Pure Pursuit	Motor signals
10	Feedback Loop	Odometry + IMU + camera	Pose update, map update, re-plan	Updated system state

3.13 Simulation Environment

The proposed system is implemented and evaluated entirely within a ROS Gazebo simulation environment. Gazebo provides a physics-accurate 3D world with configurable obstacle layouts, surface textures, and lighting conditions. ROS Noetic serves as the middleware for inter-module communication, sensor data publishing, and control command routing.Lighting conditions are varied programmatically across three levels bright, normal, and dim to assess system performance under illumination changes. Dynamic obstacles are introduced as moving entities within the simulation world to evaluate re-planning and collision avoidance under non-static conditions.

3.14 Conclusion

The proposed methodology addresses the identified research gaps through a unified, RGB only pipeline that integrates monocular depth estimation, depth-aware obstacle analysis, and closed loop navigation within a reproducible ROS Gazebo simulation framework. The system is designed to operate under varied lighting conditions and dynamic obstacle environments, and is evaluated using both collision rate and navigation success rate as primary performance metrics. The experimental setup and evaluation protocol are described in Chapter 4.

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